



Green Skills in SMEs: Definition, Relevance and Strategic Importance

🕒 15 min · Supporting learning resource (LSM)

Module 2 · From Awareness to Action-Skills for Sustainable Business

Unit 1 · Green Skills in SMEs: Definition, Relevance and Strategic Importance

→ Introduction

The transition towards a sustainable and climate-neutral economy is no longer a future objective but an immediate strategic priority for businesses. Within the European context, policy frameworks such as the **European Green Deal** and increasing regulatory mechanisms (e.g., ESG reporting, CSRD) are redefining how organizations operate.

Small and Medium-sized Enterprises (SMEs), which constitute approximately **99% of all businesses in the European Union**, are central actors in this transition. However, due to structural limitations—such as restricted financial capacity, limited human resources, and lower participation in formal training—they face disproportionate challenges in adapting to sustainability requirements.

In this context, **green skills emerge as a critical enabler**, bridging the gap between sustainability awareness and operational implementation. As highlighted in the CREDIT project materials, the key challenge for SMEs is not the lack of awareness, but the difficulty of translating this awareness into **concrete, everyday business practices**.

Key concepts explained

Conceptualising Green Skills in the SME Context

Green skills are broadly defined as the knowledge, abilities, values, and attitudes required to support a sustainable and resource-efficient society.

This definition reflects a shift from viewing sustainability as a specialized domain to understanding it as a cross-cutting competence embedded in all job roles.

Importantly, green skills are not limited to technical expertise. According to the CREDIT framework and related materials, they encompass three interrelated dimensions:

- Technical dimension: operational practices such as energy management, waste reduction, and resource efficiency
- Managerial dimension: strategic decision-making, sustainable business modelling, and supply chain management
- Transversal dimension: cognitive and behavioural capabilities such as systems thinking, adaptability, and problem-solving

Knowledge, Skills and Competences: A Functional Distinction

A critical analytical distinction in sustainability education is between knowledge, skills, and competences, as emphasized in both European frameworks (e.g., GreenComp) and the CREDIT toolkit.

- Knowledge refers to the theoretical understanding of sustainability concepts (e.g., circular economy, carbon footprint).
- Skills involve the practical application of this knowledge in specific tasks (e.g., conducting a waste audit or monitoring energy consumption).
- Competences represent the autonomous and context-sensitive application of knowledge and skills to achieve measurable sustainability outcomes.

This distinction is essential because the green transition is not achieved through awareness alone. Instead, it requires the development of competences that enable individuals to act effectively within complex and dynamic workplace environments.

From Compliance to Competitiveness

Green skills are often initially perceived as a response to regulatory pressure. However, evidence suggests that their role extends far beyond compliance.

SMEs that invest in sustainability competences can:

- anticipate regulatory changes
- reduce operational risks
- access new markets and funding opportunities
- enhance their reputation and brand value

Empirical data from European SME surveys indicate that organizations adopting ESG approaches report:

- improved environmental performance (82%)
- increased customer satisfaction (75%)
- enhanced stakeholder reputation (64%)

This demonstrates that green skills function as a driver of competitiveness and resilience, rather than merely a compliance mechanism.

Green Skills as a Driver of Operational Efficiency

From an operational perspective, green skills enable SMEs to identify and eliminate “hidden inefficiencies” in resource use.

As highlighted in the module materials, inefficiencies often remain invisible because they are embedded in routine practices—such as idle energy consumption, excessive material use, or inefficient workflows.

By developing skills such as:

- energy literacy
- resource monitoring
- process optimisation

SMEs can significantly reduce costs while improving environmental performance.

The Green Skills Gap in SMEs

Despite their importance, there is a persistent skills gap in the SME sector.

Research findings indicate that:

- SMEs have lower participation in training compared to larger enterprises
- many lack dedicated roles for sustainability
- there is a tendency to underestimate internal inefficiencies

This gap creates multiple risks:

- Efficiency risk → higher operational costs
- Market risk → exclusion from sustainable supply chains
- Innovation risk → inability to respond to emerging green markets

The CREDIT framework emphasizes that addressing this gap requires contextualized, work-based learning, rather than purely theoretical training.

Barriers to the Development of Green Skills

The adoption of green skills in SMEs is constrained by both structural and behavioural factors.

Structural Barriers

- Limited financial resources
- Lack of time for training
- Absence of specialized staff
- Complexity of regulatory frameworks

Behavioural and Organizational Barriers

- Resistance to change
- Low perceived relevance of sustainability
- Lack of incentives
- Weak organizational culture around sustainability

These barriers highlight that the green transition is not only a technical challenge but also a behavioural and organizational transformation process.

From Awareness to Action: The Core Challenge

One of the central insights emerging from the module materials is that the main barrier to sustainability is not awareness, but implementation. Many SMEs are aware of environmental issues but struggle to translate this awareness into consistent practices.

This gap between “knowing” and “doing” can be addressed through:

- practical training approaches
- integration of sustainability into daily routines
- small, incremental changes (micro-actions)
- employee engagement and ownership

As highlighted in the training materials, sustainable transformation in SMEs is achieved through the accumulation of small, consistent actions, rather than large-scale interventions.

Conclusion

Green skills represent a fundamental component of the transition towards sustainable SMEs. They enable organizations to move beyond theoretical awareness and develop the competences necessary for real-world implementation.

For SMEs, the development of green skills is not only an environmental necessity but also a strategic opportunity to enhance efficiency, competitiveness, and resilience. Ultimately, the green transition depends on the ability of individuals and organizations to continuously learn, adapt, and integrate sustainability into everyday decision-making and operations.

Bibliography / References

- European Commission (2020). The European Green Deal
- European Commission (2023). SME Sustainability Report (SME EnterPRIZE)
- Cedefop (2021). Skills for Green Jobs
- OECD (2022). SMEs and the Green Transition
- UNESCO-UNEVOC (2021). Greening Technical and Vocational Education and Training (TVET)
- European Commission (2022). GreenComp: The European Sustainability Competence Framework
- CREDIT Project Consortium (2025). Module 2 Training Materials
- “Green Skills for SMEs in Real World Actions”
- “Module 2 Unit 1 – Awareness to Action Framework”
- “Green Skills: Key Concepts and Categories”